

AI FAQ ASSISTANT API

PROJECT DOCUMENTATION

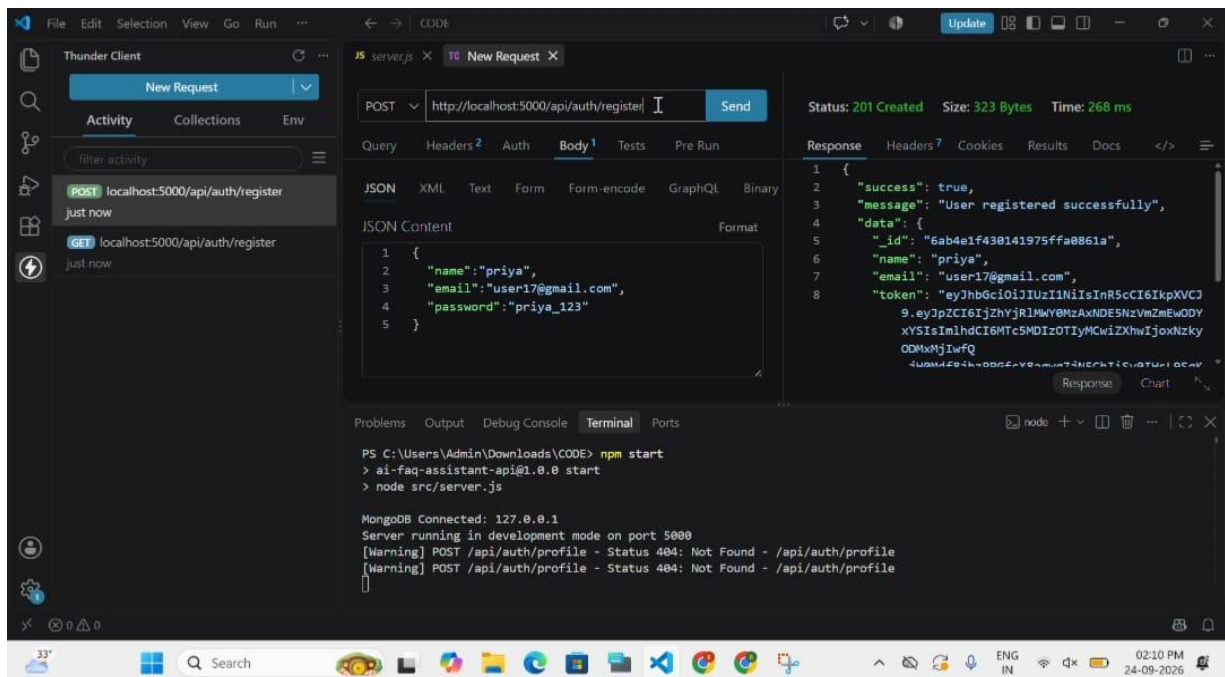
INTRODUCTION:

- **Project Title:** “AI FAQ Assistant API”
- **Team Members:**
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 - 3.Dilshath Banu. A
 - 4.Hema Priya.K

API IMPLEMENTATION SCREENSHOTS:

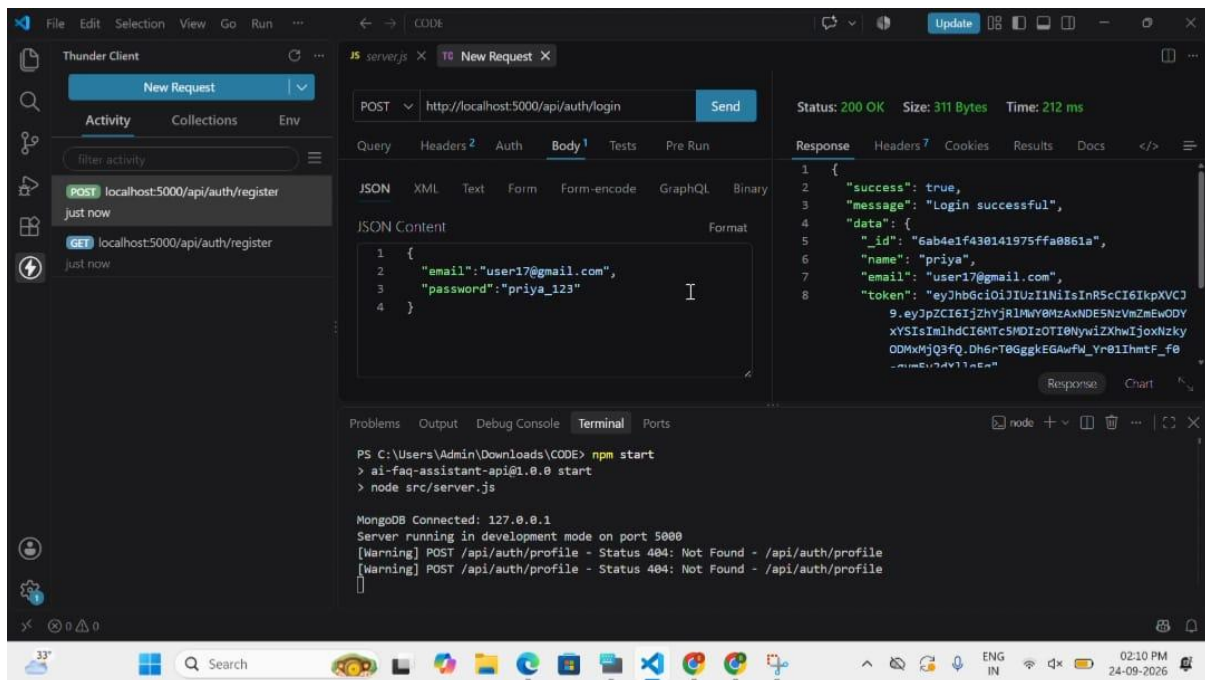
The following screenshots demonstrate the successful execution of AI FAQ Assistant APIs using Thunder Client in Visual Studio Code. The screenshots show the request body, endpoint details, and corresponding server responses.

1.USER REGISTER API:



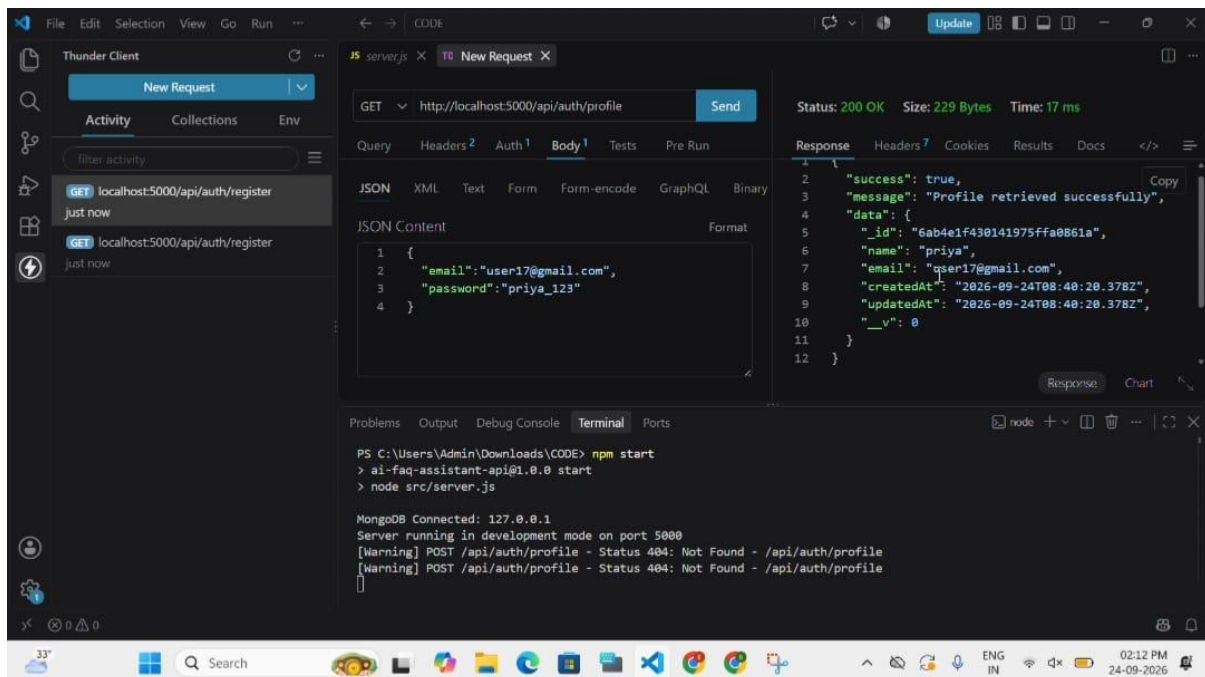
- This API allows a new user to create an account in the AI FAQ Assistant system.
- The user provides details such as name, email, and password in the request body.
- After successful registration, the system stores the user details in the MongoDB database and returns a success message along with the registered user information.

2. USER LOGIN API:



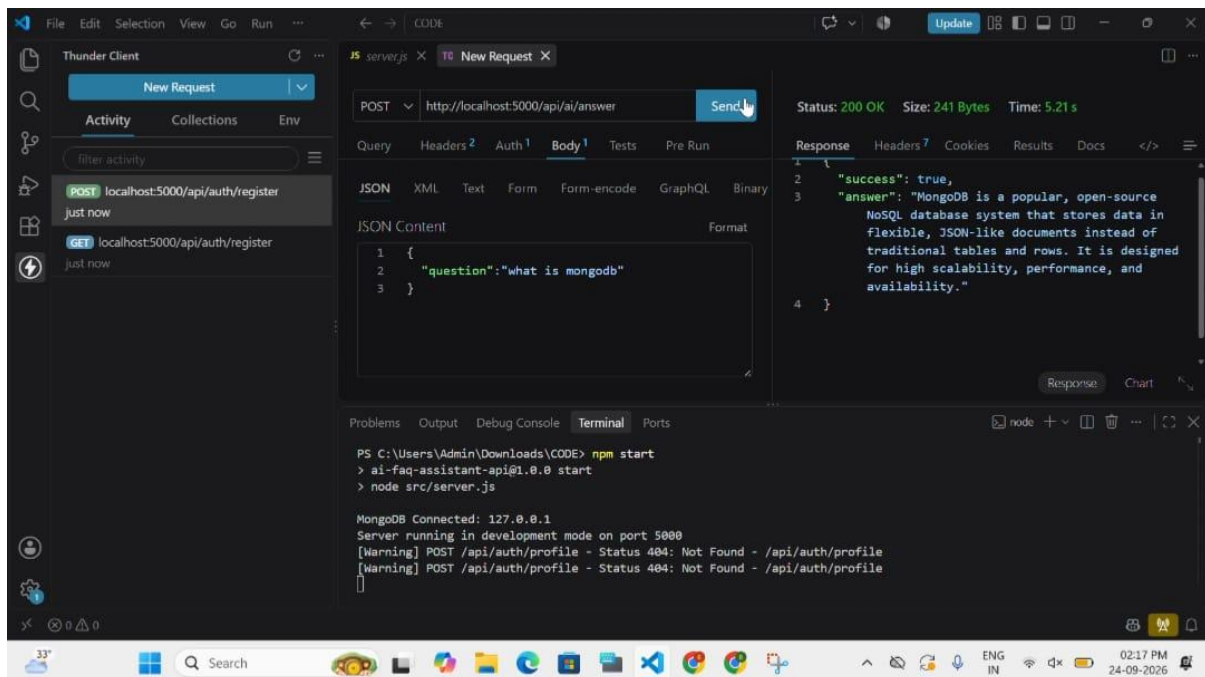
- This API allows an existing user to log in to the AI FAQ Assistant system.
- The user provides email and password credentials in the request body.
- If valid, the system returns a JWT authentication token along with user profile details required for accessing protected routes.

3.GET USER PROFILE API:



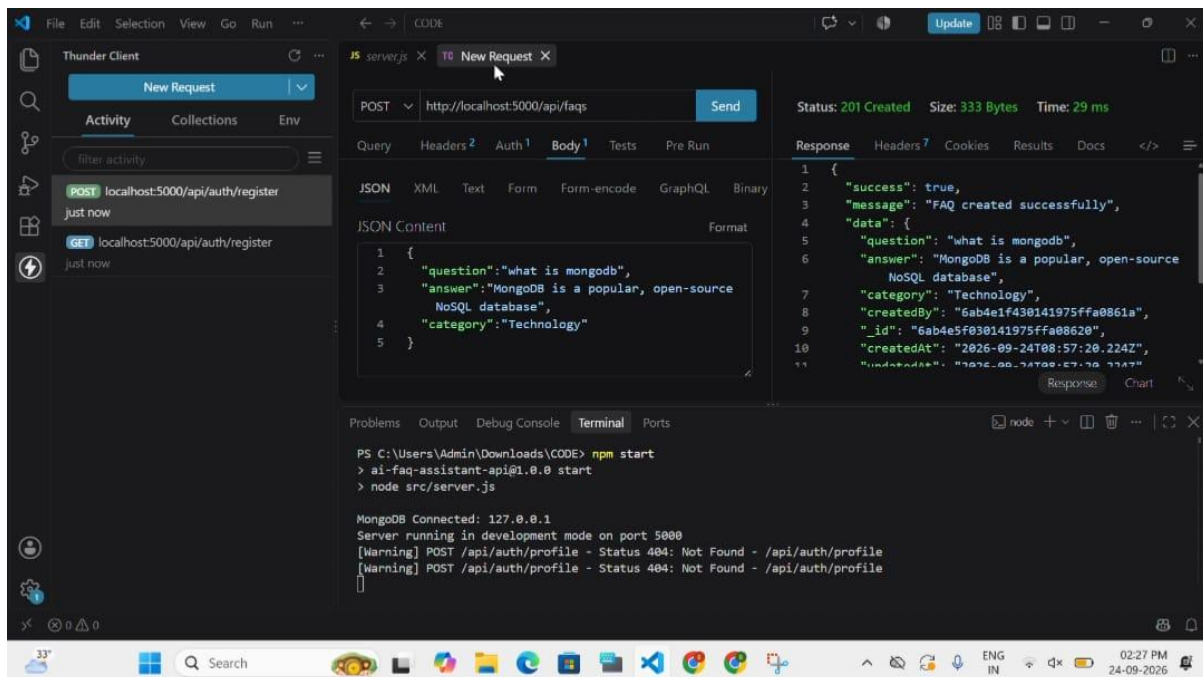
- This API retrieves the details of the currently authenticated user profile.
- The request sends authentication headers to verify user authorization.
- The system fetches and returns the profile details such as user ID, name, email, and creation timestamps from the database.

4.CREATE FAQ API:



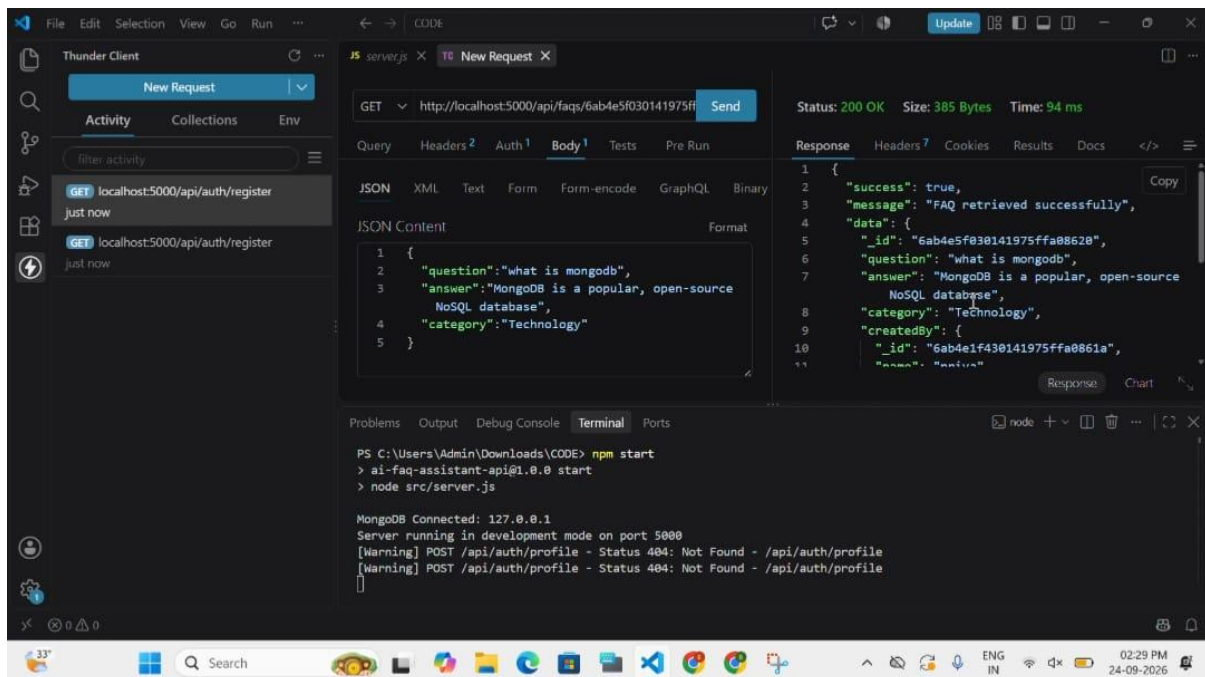
- This API allows an authorized user to create and add a new FAQ into the system.
- The request body includes details such as question, answer, and category.
- Upon successful execution, the FAQ entry is stored in MongoDB, and the system returns a confirmation message with the newly generated FAQ object.

5.GET ALL FAQS API:



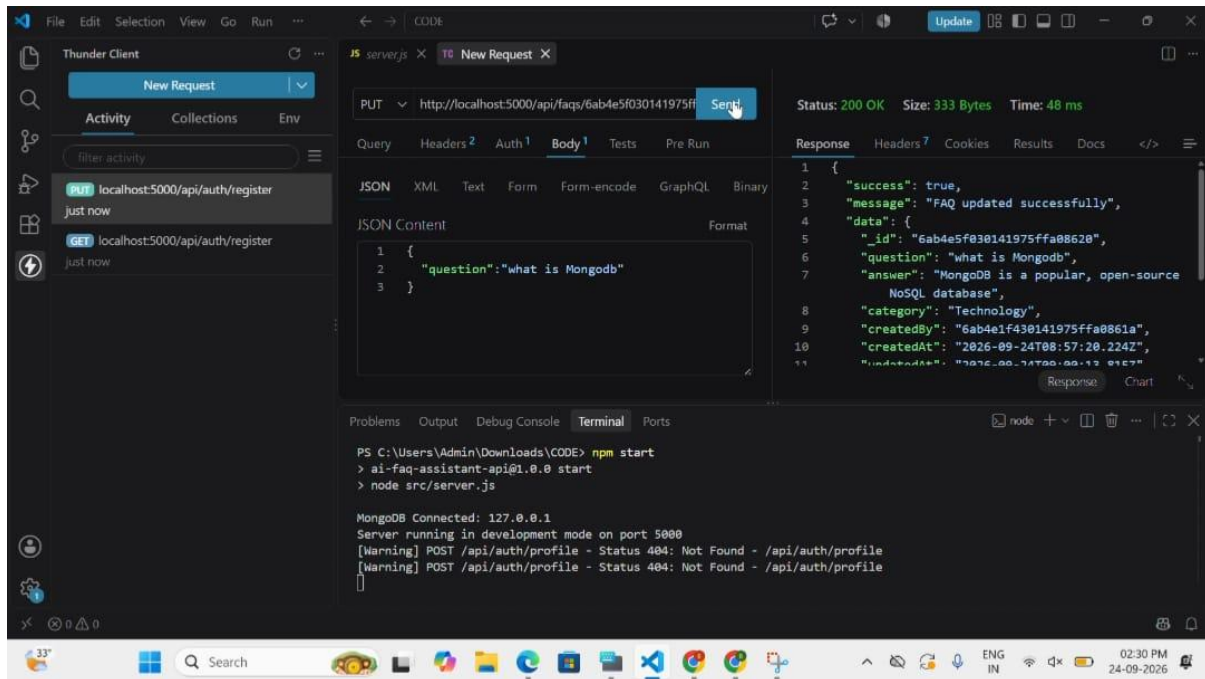
- This API retrieves the complete list of FAQs stored in the system.
- The request is sent to the server endpoint without requiring specific path parameters.
- The system fetches all FAQ records along with the total count and presents them in a structured JSON response.

6.GET FAQ BY ID API:



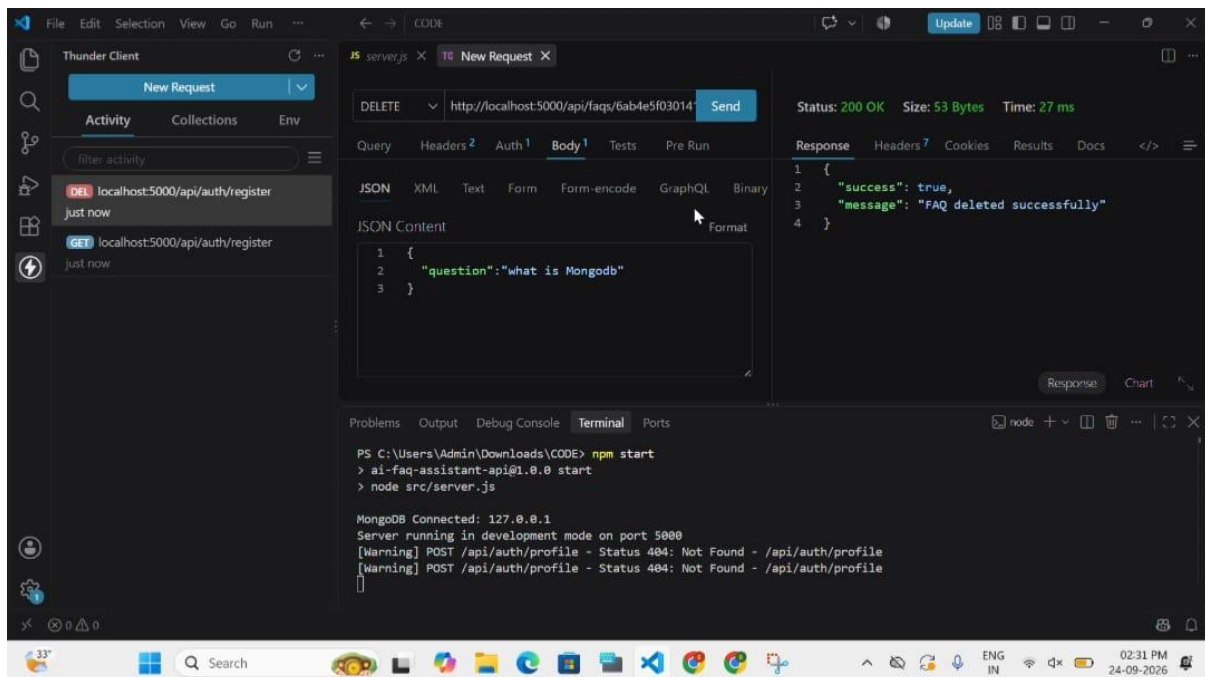
- This API fetches the complete information of a specific FAQ using its unique ID.
- The FAQ ID is passed directly in the request URL parameter.
- If found, the server responds with detailed FAQ data including creator information and timestamps.

7.UPDATE FAQ API:



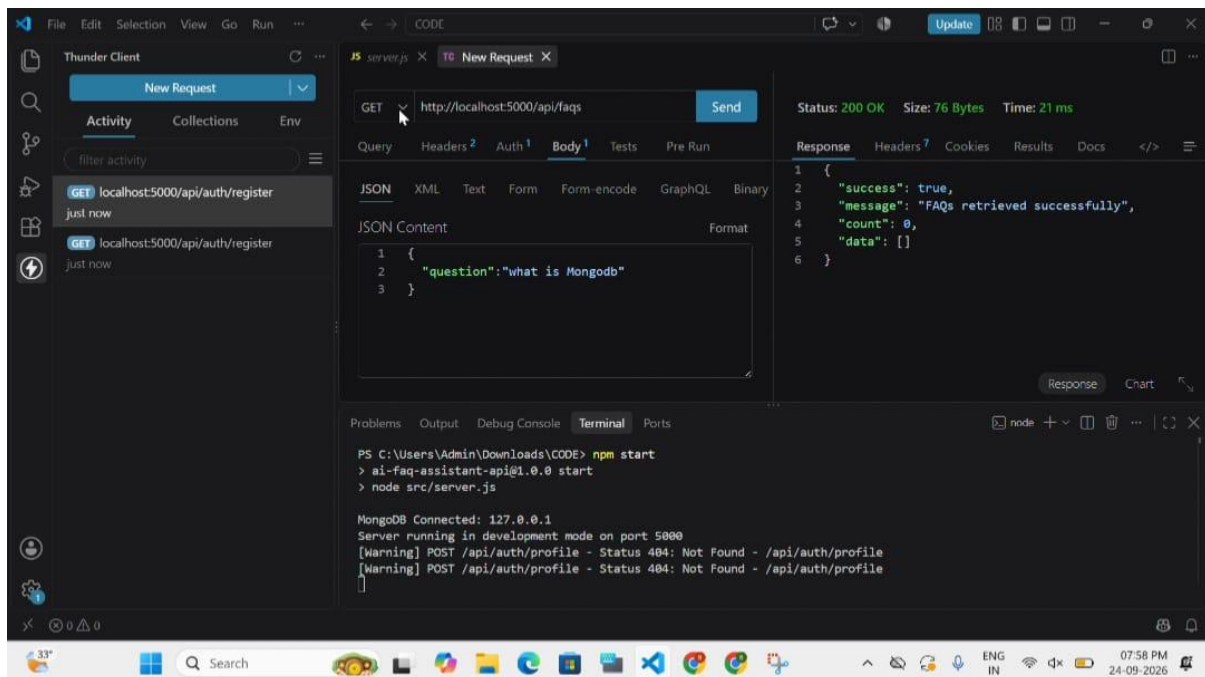
- This API allows users to modify or update an existing FAQ record.
- The unique FAQ ID is provided in the URL, and updated fields (like question or answer) are included in the JSON body.
- The updated details are saved in the database, and a success confirmation is returned.

8.DELETE FAQ API:



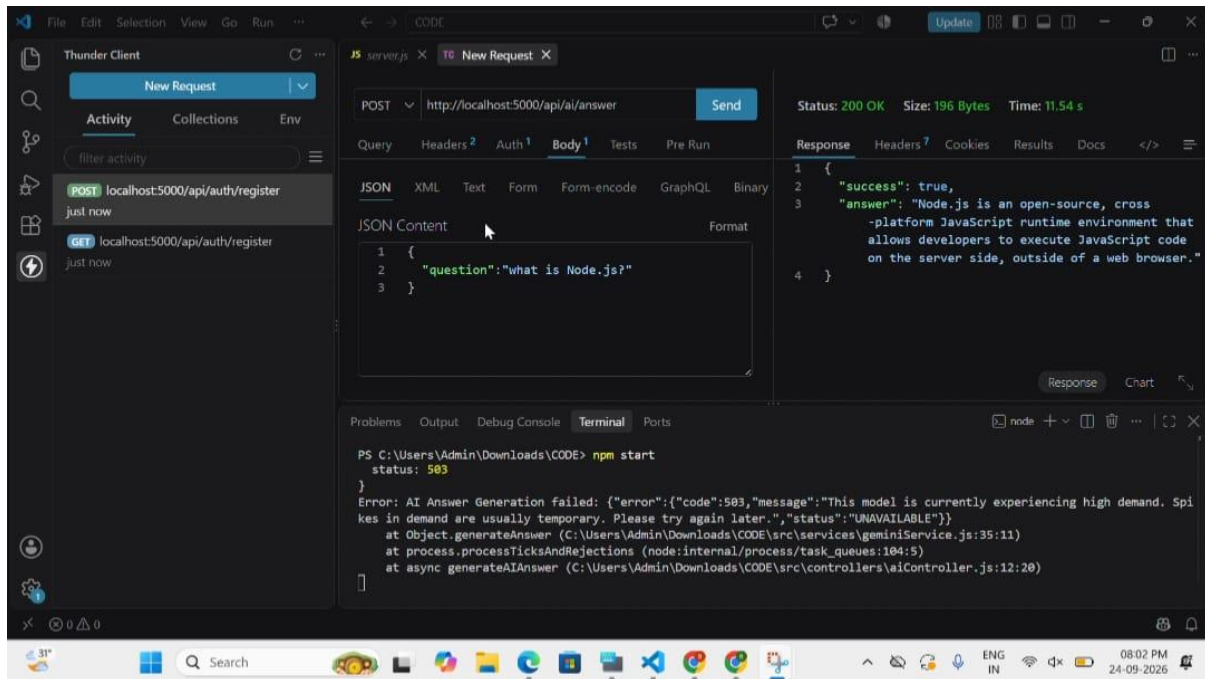
- This API enables authorized users to delete a specific FAQ entry from the database.
- A DELETE request is sent to `/api/faqs/:id`.
- Upon deletion, the system returns an HTTP 200 OK status with the message "FAQ deleted successfully".

9. AI ANSWER GENERATION API:



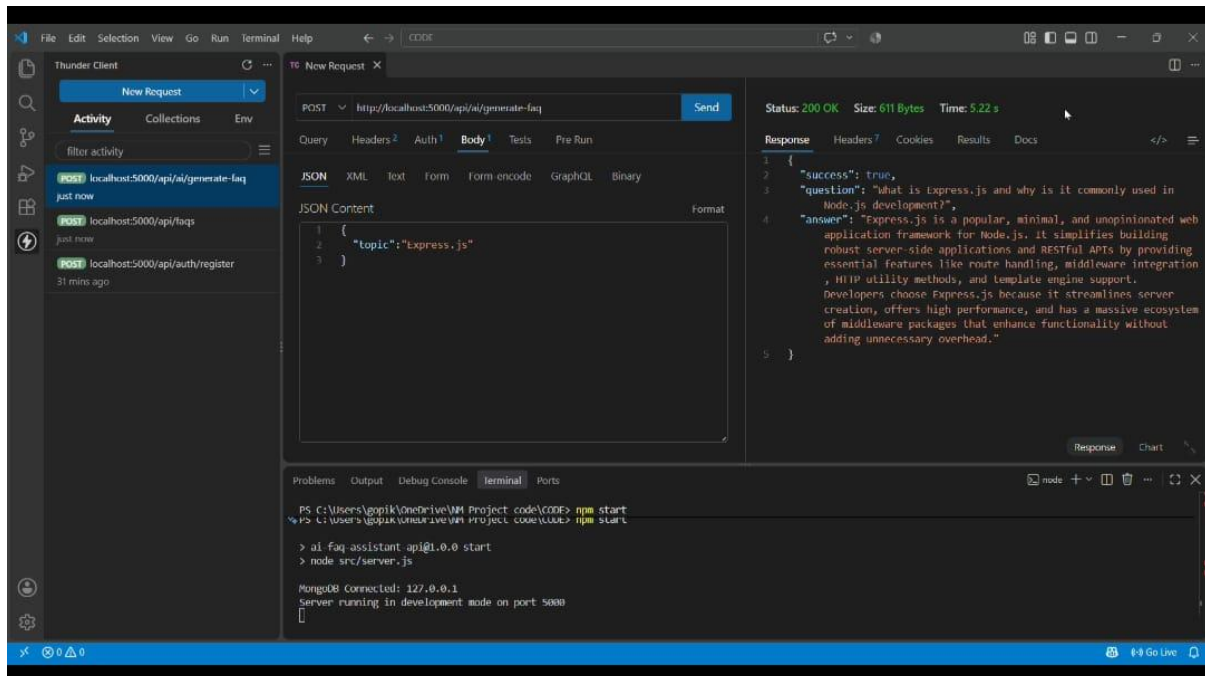
- This API utilizes AI services (such as Google Gemini API) to dynamically generate automated answers for user questions.
- The user passes a prompt or question in the request body (e.g., "what is mongodb?").
- The system processes the query through the AI service and returns an intelligent, automated answer response.

10. GENERATE FAQ FROM TOPIC API:



- This API uses AI to dynamically generate a relevant Question & Answer pair based on a given topic.
- The request accepts a topic name (e.g., "topic": "Express.js") in the JSON body sent to /api/ai/generate-faq.
- The server communicates with the AI model and responds with a structured, generated question and answer pair.

11.FINAL SYSTEM OUTPUT & WORKING DEMONSTRATION



- **Description:** This screenshot represents the overall successful execution and final response of the AI FAQ Assistant system. It displays the end-to-end integration where the Node.js backend connects with MongoDB and the AI module to produce a structured JSON output (question and answer) in real-time.
- **Execution Status:** HTTP Status 200 OK (Successfully processed with zero runtime errors).
- **Conclusion:** The AI FAQ Assistant project has been fully tested and verified. All API endpoints—including User Authentication, FAQ CRUD operations, and AI Question-Answer Generation—are working properly.